

Here are your **1-time-read, exam-ready notes** — everything compressed, simple, and complete:

Information Security — Quick Revision Notes

Cyber Security

- Protection of **data during storage, transmission, and processing**
 - Covers: computers, networks, devices, Internet
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Information Security (Subset of Cyber Security)

Focus: **CIA + extra properties**

Core (CIA Triad)

- **Confidentiality** → No unauthorized access
- **Integrity** → No unauthorized modification
- **Availability** → System/data accessible when needed

Additional:

- **Authenticity** → Data/source is genuine
 - **Accountability** → Actions traceable to entity
 - **Non-repudiation** → Cannot deny action
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Basic Definitions

- **Computer Security** → Protect data + stop hackers
 - **Network Security** → Protect data during transmission
 - **Internet Security** → Protection over interconnected networks
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Security Objectives (Important)

1. Confidentiality

- **Data confidentiality** → protect data
- **Privacy** → user controls personal data

2. Integrity

- **Data integrity** → data unchanged
- **System integrity** → system works correctly

3. Availability

- System/services always accessible

4. Authenticity

- Verify identity/source

5. Accountability

- Actions traceable
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Computer Security Challenges

- Complex, not simple
 - Need to anticipate attacks
 - Uses **algorithms + secrets**
 - Continuous monitoring required
 - Often ignored until failure
 - Trade-off: security vs usability
 - Ongoing battle: **attacker vs defender**
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OSI Security Architecture

- Standard: **ITU-T X.800**
 - Provides structured way to define:
 - attacks
 - services
 - mechanisms
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3 Aspects of Security

1. **Security Attack**
2. **Security Mechanism**
3. **Security Service**

Security Attack

- Any action that **breaks security**

Types:

- **Passive Attack**
 - Monitor only (no change)
 - Example: eavesdropping
- **Active Attack**
 - Modify/disrupt data
 - Example: data tampering

Security Services

 Goal: **Protect systems & data**

Types:

- **Authentication** → verify identity
- **Access Control** → restrict access
- **Data Confidentiality** → hide data
- **Data Integrity** → ensure no change
- **Non-repudiation** → no denial

Security Mechanisms

 Tools to **detect/prevent/recover attacks**

Types:

1. Specific Mechanisms

- Encryption (encipherment)
- Digital signatures
- Access control
- Authentication exchange
- Traffic padding

- Routing control
- Notarization

2. Pervasive Mechanisms

- Trusted systems
 - Security labels
 - Event detection
 - Audit trails
 - Recovery
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Cryptography

- Mathematical technique to **transform data**
 - Used for:
 - secure storage
 - secure communication
 - authentication
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Cryptographic Algorithms

1. Keyless

- **Hash Function**
 - Converts data → fixed output
 - Used in signatures, MAC
 - **PRNG**
 - Generates pseudo-random numbers
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2. Single-Key (Symmetric)

- Same key for encryption/decryption

Types:

- **Block Cipher**
 - Works on blocks (e.g., 128 bits)

- **Stream Cipher**
 - Works bit-by-bit (uses XOR)
 - **MAC**
 - Ensures integrity + authenticity
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3. Two-Key (Asymmetric)

- Uses **public + private key**

Includes:

- **Digital Signature**
 - Verifies **origin + integrity**
 - **Key Exchange**
 - Secure key sharing
 - **User Authentication**
 - Verifies user identity
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Final Exam Tips (Important)

- **CIA Triad → MUST remember**
 - Difference:
 - **Attack vs Service vs Mechanism**
 - **Passive vs Active attacks**
 - **Symmetric vs Asymmetric**
 - Cryptography → core concept
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If you send next lecture, I'll compress it the same way.

Here is your **Lecture 2: Classical Encryption Techniques** — ultra-compressed, exam-ready notes 📌

🔒 Classical Encryption Techniques

📌 Symmetric Encryption (IMPORTANT)

- **Plaintext** → original message
 - **Ciphertext** → encrypted message
 - **Encryption** → plaintext → ciphertext
 - **Decryption** → ciphertext → plaintext
 - **Cryptanalysis** → breaking cipher without key
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📌 Symmetric Cipher Model

3 components:

- **Encryption Algorithm** → transforms data
- **Secret Key** → controls encryption
- **Decryption Algorithm** → reverse process

👉 Same key used → **Symmetric**

📌 Attacks on Encryption

1. Cryptanalysis

- Uses:
 - algorithm knowledge
 - plaintext patterns
- Smarter attack

2. Brute Force

- Try all possible keys
 - Avg tries = **half of key space**
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📌 Types of Techniques

👉 2 building blocks:

- **Substitution** → replace characters
 - **Transposition** → rearrange characters
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🔐 Substitution Techniques

📌 Caesar Cipher (VERY IMPORTANT)

- Shift letters by **k positions**

Formula:

- **Encryption:** $C = (P + k) \bmod 26$
- **Decryption:** $P = (C - k) \bmod 26$

Example:

- meet → PHHW (shift = 3)

Why weak?

- Only **25 keys**
 - Easy brute-force
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📌 Monoalphabetic Cipher

- Each letter → fixed substitution
- Example:
 - TRY → GBW

Key Space:

- **26!** ($\sim 10^{26}$) → very large

Weakness:

- **Frequency analysis**
 - English letters pattern (e, t, a...)

Countermeasure:

- **Homophones** → multiple symbols per letter
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🔐 Polyalphabetic Ciphers

👉 Improvement over monoalphabetic

Features:

- Multiple substitution rules
 - Key decides which rule
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📌 Vigenère Cipher (VERY IMPORTANT)

- Uses **multiple Caesar ciphers (0–25 shifts)**
- Key is **repeated**

Example:

- Key: deceptive
 - Output: ZICVTWQNGRZGVTWAVZHCQYGLMGJ
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📌 Vigenère Cryptanalysis

- Find **key length**
 - Use **repeating patterns**
 - Use **frequency analysis**
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📌 Autokey Cipher

- Key = **keyword + plaintext**
 - Removes repetition → more secure
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📌 Vernam Cipher

- Uses **bitwise operation (XOR)**
 - Foundation of **One-Time Pad**
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🧠 Key Comparisons (EXAM GOLD)

Technique	Security Weakness
Caesar	Very Low 25 keys

Technique	Security Weakness	
Monoalphabetic	Medium	Frequency attack
Vigenère	Higher	Pattern detection
Autokey	Better	Still breakable
Vernam	Strong	Key handling

⚠ Important Concepts

- **Frequency Analysis** → main attack
 - **Key Space** → number of possible keys
 - **Brute Force vs Cryptanalysis**
 - **Substitution vs Polyalphabetic**
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🎯 Final Quick Recall

- Caesar → shift
 - Monoalphabetic → fixed mapping
 - Vigenère → repeating key
 - Autokey → key + plaintext
 - Vernam → XOR
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Send next lecture 👍

Here are your **targeted notes (from One-Time Pad → Rotor Ciphers)** — fully compressed, nothing skipped:

One-Time Pad (OTP) — VERY IMPORTANT

Concept

- Proposed by **Joseph Mauborgne**
- Key = **random + same length as message**
- Used **only once** → then discarded

Properties

- Produces **completely random ciphertext**
- **Unbreakable (perfect security)**
- No statistical relation with plaintext

Why perfectly secure?

- Every possible plaintext equally likely
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Practical Problems

-  Generating truly random keys (huge amount)
-  Key distribution problem
 - Sender & receiver need same long key
-  Key management is very difficult

 **Conclusion:** Theoretically perfect, practically hard

Transposition Cipher

Concept

- Rearrange positions of letters (no substitution)
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Types

1. Rail Fence Cipher

- Write in zig-zag (rows)
 - Example:
 - meet me after → rearranged output
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2. Columnar Transposition

- Write in rows
 - Read column-wise (with permutation)
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Multiple Transposition

- Apply cipher multiple times
 - More secure than single transposition
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Key Point

- Keeps **same letters**, only changes **order**
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Product Ciphers (VERY IMPORTANT)

Concept

- Combine multiple ciphers

Cases:

- Substitution + substitution → still substitution
- Transposition + transposition → still transposition
-  **Substitution + Transposition → Strong cipher**

 This idea leads to **modern encryption (DES, AES)**

Rotor Ciphers

Concept

- Pre-modern **complex encryption machines**
- Used in **World War II**

Examples

- German **Enigma**
- Allied **Hagelin**
- Japanese **Purple**

Working

- Uses **rotating cylinders (rotors)**
- Each rotor = substitution
- After each letter:
 - Rotor rotates → mapping changes

 Result: **Different encryption for each letter**

Key Idea

- Dynamic substitution (not fixed like monoalphabetic)

Key Space

- With 3 rotors:
 - **$26^3 = 17,576$ combinations**

Advantages

- Much stronger than classical ciphers
- Hard to break (at that time)

Limitation

- Eventually broken (e.g., Enigma in WWII)

Final Quick Recall

- OTP → perfect but impractical
- Transposition → rearrangement
- Product cipher → combination = strong
- Rotor → rotating substitution (dynamic)

If you want, I can give you **expected exam questions from this lecture** (very helpful for last-day prep).